# Derivation compendium and audit notes

corrected companion for the Laplace-focused presentation

March 2026

This companion records the main corrections to the original draft and proves every formula used in the short presentation note. The emphasis is on explicit algebra: products of Gaussians are completed by hand, matrix identities are shown line by line, and every derivative used in the $K = 1$ Laplace benchmark is written out.

## Contents

# 1 Scope and main corrections

The clean benchmark retained in the final short note is the two-frame Laplace AR(1) model

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \qquad A_1 = \alpha A_0 + \varepsilon, \qquad \varepsilon \sim \mathrm{Laplace}(0, b), \qquad |\alpha| < 1, \tag{1}$$

followed by OU noising

$$X_k = e^{-t} A_k + \sqrt{\Delta_t}\, Z_k, \qquad \Delta_t = 1 - e^{-2t}, \qquad Z_0, Z_1 \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{2}$$

The original draft contained several mathematical issues. The final package corrects them as follows.

1. The notation $a_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ was inconsistent: for the scalar initial variable one needs $a_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$, while $\Sigma_0$ denotes the covariance of the full trajectory vector.
2. The displayed finite-chain Gaussian AR(1) precision matrix was not exact at the boundaries. The first and last diagonal entries differ from the interior entries; the correct matrix is derived in Section 2.1.
3. The locality diagnostics based on the full Frobenius norm of $Q_t$ were misleading at large time, because $Q_t \to I$. The corrected diagnostics normalize only off-diagonal mass; see Section 2.4.
4. The deterministic control was mathematically correct but conceptually overinterpreted. Its dense noised precision comes from a rank-one propagation mechanism, not from the same Markov fill-in mechanism as the stochastic AR(1) chain; see Section 2.5.
5. The old normal-Laplace convolution formula used in the code was wrong. The corrected formula is derived in Section 4.
6. The old $K = 2$ Laplace "closed-form" claim is not retained. What is justified is an exact nested integral representation, not an elementary closed form comparable to the $K = 1$ formula; see Section 7.

The rest of the document proves the corrected statements explicitly.

# 2 Gaussian and deterministic audit

## 2.1 Exact finite-chain Gaussian AR(1) precision

Consider the Gaussian AR(1) chain

$$A_{k+1} = \alpha A_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2), \qquad k = 0, \ldots, K-1, \tag{3}$$

with

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2). \tag{4}$$

Its joint density is

$$p(a_0, \dots, a_K) \propto \exp\left( -\frac{(a_0 - \mu_0)^2}{2\sigma_0^2} - \sum_{k=0}^{K-1} \frac{(a_{k+1} - \alpha a_k)^2}{2\sigma_\eta^2} \right). \tag{5}$$

We expand the quadratic form in (5) exactly.

**Step 1: expand the initial term.**

$$\frac{(a_0 - \mu_0)^2}{2\sigma_0^2} = \frac{a_0^2}{2\sigma_0^2} - \frac{\mu_0}{\sigma_0^2} a_0 + \frac{\mu_0^2}{2\sigma_0^2}. \tag{6}$$

**Step 2: expand each innovation term.** For each $k$,

$$\frac{(a_{k+1} - \alpha a_k)^2}{2\sigma_\eta^2} = \frac{a_{k+1}^2}{2\sigma_\eta^2} - \frac{\alpha}{\sigma_\eta^2} a_k a_{k+1} + \frac{\alpha^2 a_k^2}{2\sigma_\eta^2}. \tag{7}$$

**Step 3: collect the coefficients of each $a_j^2$.**

- The variable $a_0$ appears in (6) and in the $k = 0$ instance of (7). Therefore its quadratic coefficient is
$$\frac{1}{2\sigma_0^2} + \frac{\alpha^2}{2\sigma_\eta^2}.$$
- An interior variable $a_j$, with $1 \le j \le K - 1$, appears
  - as $a_{k+1}$ when $k = j - 1$, giving $a_j^2/(2\sigma_\eta^2)$,
  - as $a_k$ when $k = j$, giving $\alpha^2 a_j^2/(2\sigma_\eta^2)$.

  Therefore its quadratic coefficient is
$$\frac{1 + \alpha^2}{2\sigma_\eta^2}.$$
- The last variable $a_K$ appears only as $a_{k+1}$ when $k = K - 1$. Therefore its quadratic coefficient is
$$\frac{1}{2\sigma_\eta^2}.$$

**Step 4: collect cross terms.** From (7), the only cross term involving $a_k$ and $a_{k+1}$ is

$$-\frac{\alpha}{\sigma_\eta^2} a_k a_{k+1}. \tag{8}$$

No other cross terms appear. Hence the precision matrix is tridiagonal.

Putting these pieces together,

$$p(a) \propto \exp\left( -\frac{1}{2} a^\top Q_0 a + h^\top a \right), \tag{9}$$

with precision

$$Q_0 = \begin{pmatrix} \sigma_0^{-2} + \alpha^2 \sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} & & & \\ -\alpha\sigma_\eta^{-2} & (1+\alpha^2)\sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} & & \\ & \ddots & \ddots & \ddots & \\ & & -\alpha\sigma_\eta^{-2} & (1+\alpha^2)\sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} \\ & & & -\alpha\sigma_\eta^{-2} & \sigma_\eta^{-2} \end{pmatrix}. \tag{10}$$

**Stationary specialization.** If $A_0$ is chosen with stationary variance

$$\sigma_0^2 = \frac{\sigma_\eta^2}{1-\alpha^2}, \tag{11}$$

then

$$\sigma_0^{-2} + \alpha^2\sigma_\eta^{-2} = \frac{1-\alpha^2}{\sigma_\eta^2} + \frac{\alpha^2}{\sigma_\eta^2} = \frac{1}{\sigma_\eta^2}.$$

Hence the first and last diagonal entries coincide, and

$$Q_0 = \frac{1}{\sigma_\eta^2}\begin{pmatrix} 1 & -\alpha & & & \\ -\alpha & 1+\alpha^2 & -\alpha & & \\ & \ddots & \ddots & \ddots & \\ & & -\alpha & 1+\alpha^2 & -\alpha \\ & & & -\alpha & 1 \end{pmatrix}. \tag{12}$$

This is the correct finite-chain formula. The original short draft displayed the interior pattern on every diagonal entry, which is not exact at the boundaries.

## 2.2 OU noising and the exact Gaussian precision identities

Let $A \in \mathbb{R}^n$ be any random vector with mean $\mu_0$ and covariance $\Sigma_0$. Define

$$X_t = e^{-t}A + \sqrt{\Delta_t}\, Z, \qquad Z \sim \mathcal{N}(0, I_n), \qquad Z \perp A. \tag{13}$$

Then

$$\mathbb{E}[X_t] = e^{-t}\mathbb{E}[A] + \sqrt{\Delta_t}\,\mathbb{E}[Z] = e^{-t}\mu_0, \tag{14}$$

$$\begin{aligned} \mathrm{Cov}(X_t) &= \mathrm{Cov}(e^{-t}A + \sqrt{\Delta_t}Z) \\ &= e^{-2t}\mathrm{Cov}(A) + \Delta_t\mathrm{Cov}(Z) + e^{-t}\sqrt{\Delta_t}\,\mathrm{Cov}(A,Z) + e^{-t}\sqrt{\Delta_t}\,\mathrm{Cov}(Z,A). \end{aligned} \tag{15}$$

Because $A$ and $Z$ are independent, $\mathrm{Cov}(A,Z) = \mathrm{Cov}(Z,A) = 0$, and because $\mathrm{Cov}(Z) = I_n$, (15) becomes

$$\Sigma_t := \mathrm{Cov}(X_t) = e^{-2t}\Sigma_0 + \Delta_t I_n. \tag{16}$$

**SNR form of the precision.** Let

$$\gamma_t := \frac{e^{-2t}}{\Delta_t}. \tag{17}$$

Then

$$\Sigma_t = \Delta_t(I_n + \gamma_t\Sigma_0),$$

so

$$Q_t := \Sigma_t^{-1} = \Delta_t^{-1}(I_n + \gamma_t\Sigma_0)^{-1}. \tag{18}$$

**Factorization through $Q_0 = \Sigma_0^{-1}$.** Since $\Delta_t = 1 - e^{-2t}$, one has

$$e^{2t}\Delta_t = e^{2t} - 1 =: c_t. \tag{19}$$

Rewrite (16) as

$$\Sigma_t = e^{-2t}(\Sigma_0 + c_t I_n). \tag{20}$$

Invert (20):

$$Q_t = e^{2t}(\Sigma_0 + c_t I_n)^{-1}. \tag{21}$$

Now use $\Sigma_0 = Q_0^{-1}$:

$$\Sigma_0 + c_t I_n = Q_0^{-1} + c_t I_n = Q_0^{-1}(I_n + c_t Q_0).$$

Hence

$$(\Sigma_0 + c_t I_n)^{-1} = (I_n + c_t Q_0)^{-1} Q_0.$$

Substitute into (21):

$$Q_t = e^{2t}(I_n + c_t Q_0)^{-1} Q_0. \tag{22}$$

Because $Q_0$ commutes with every polynomial in $Q_0$, it also commutes with $(I_n + c_t Q_0)^{-1}$. Therefore

$$Q_t = e^{2t} Q_0 (I_n + c_t Q_0)^{-1}. \tag{23}$$

**Neumann expansion.** If $c_t \|Q_0\| < 1$ in any submultiplicative operator norm, then

$$(I_n + c_t Q_0)^{-1} = \sum_{m=0}^{\infty} (-c_t)^m Q_0^m. \tag{24}$$

Insert this into (23):

$$Q_t = e^{2t} \sum_{m=0}^{\infty} (-c_t)^m Q_0^{m+1}. \tag{25}$$

**Why distance-$d$ couplings appear at order $t^{d-1}$.** If $Q_0$ is tridiagonal, then $Q_0$ has bandwidth 1: $(Q_0)_{ij} = 0$ whenever $|i-j| > 1$.

If $B$ has bandwidth $r$ and $C$ has bandwidth $s$, then $BC$ has bandwidth at most $r+s$. Indeed,

$$(BC)_{ij} = \sum_k B_{ik} C_{kj}.$$

For a term in the sum to be nonzero, one must have both $|i-k| \le r$ and $|k-j| \le s$. Therefore $|i-j| \le r+s$. Hence $Q_0^{m+1}$ has bandwidth at most $m+1$.

In particular, an entry with $|i-j| = d$ cannot appear before the term $Q_0^d$, which corresponds to $m = d-1$ in (25). For small $t$,

$$c_t = e^{2t} - 1 = 2t + O(t^2),$$

so the first nonzero contribution to a distance-$d$ entry is of order $t^{d-1}$.

**Large-time expansion.** Let $\varepsilon = e^{-2t}$. Then $\Delta_t = 1 - \varepsilon$, and (16) becomes

$$\Sigma_t = I_n + \varepsilon(\Sigma_0 - I_n).$$

For small $\varepsilon$,

$$(I_n + \varepsilon M)^{-1} = I_n - \varepsilon M + O(\varepsilon^2).$$

Apply this with $M = \Sigma_0 - I_n$:

$$Q_t = I_n - e^{-2t}(\Sigma_0 - I_n) + O(e^{-4t}). \tag{26}$$

### 2.3 Gaussian score and posterior mean

If $X_t \sim \mathcal{N}(\mu_t, \Sigma_t)$ with precision $Q_t = \Sigma_t^{-1}$, then

$$p_t^{\mathrm{G}}(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma_t)^{1/2}} \exp\Big(-\frac{1}{2}(x-\mu_t)^\top Q_t (x-\mu_t)\Big). \tag{27}$$

Take logarithms:

$$\log p_t^{\mathrm{G}}(x) = C_t - \frac{1}{2}(x-\mu_t)^\top Q_t (x-\mu_t),$$

where $C_t$ does not depend on $x$. Expand the quadratic term:

$$(x-\mu_t)^\top Q_t (x-\mu_t) = x^\top Q_t x - 2\mu_t^\top Q_t x + \mu_t^\top Q_t \mu_t.$$

Therefore

$$\log p_t^{\mathrm{G}}(x) = C_t' - \frac{1}{2} x^\top Q_t x + \mu_t^\top Q_t x.$$

Differentiate with respect to $x$:

$$\nabla \log p_t^{\mathrm{G}}(x) = -Q_t(x-\mu_t), \qquad -\nabla^2 \log p_t^{\mathrm{G}}(x) \equiv Q_t. \tag{28}$$

The posterior mean of the clean state given $X_t = x$ is the standard Gaussian regression formula. Since

$$\mathrm{Cov}(A, X_t) = \mathrm{Cov}\big(A, e^{-t}A + \sqrt{\Delta_t} Z\big) = e^{-t}\Sigma_0,$$

and $\mathrm{Cov}(X_t) = \Sigma_t$, one gets

$$\mathbb{E}[A \mid X_t = x] = \mu_A + e^{-t}\Sigma_0 Q_t (x - \mu_t). \tag{29}$$

### 2.4 Corrected locality diagnostics

The original short draft normalized band energies by $\|Q_t\|_F^2$ including the diagonal. This becomes misleading at large $t$, because $Q_t \to I$, so the diagonal dominates automatically.

A more meaningful off-diagonal normalization is

$$E_d(t) := \sum_{|i-j|=d} Q_t[i,j]^2, \qquad E_{\mathrm{off}}(t) := \sum_{d \geq 1} E_d(t), \qquad p_d(t) := \frac{E_d(t)}{E_{\mathrm{off}}(t)}. \tag{30}$$

From these one can define

$$b_{0.95}(t) := \min\left\{ r : \sum_{d=1}^{r} p_d(t) \geq 0.95 \right\}, \qquad \bar{\xi}(t) := \sum_{d \geq 1} d\, p_d(t). \tag{31}$$

These observables measure the shape of the off-diagonal coupling profile rather than the trivial approach to the identity.

### 2.5 Deterministic control

The deterministic control is

$$A_k = \alpha^k A_0, \qquad A_0 \sim \mathcal{N}(0, \sigma_0^2), \qquad v := (1, \alpha, \ldots, \alpha^K)^\top. \tag{32}$$

Then the clean covariance is rank one:

$$\Sigma_0^{\det} = \sigma_0^2 v v^\top. \tag{33}$$

After OU noising,

$$\Sigma_t^{\det} = \Delta_t I + \beta_t vv^\top, \qquad \beta_t := e^{-2t}\sigma_0^2. \tag{34}$$

We invert (34) by Sherman-Morrison. Write

$$A = \Delta_t I, \qquad u = \sqrt{\beta_t}\, v, \qquad \Sigma_t^{\det} = A + uu^\top.$$

The Sherman-Morrison formula gives

$$(A + uu^\top)^{-1} = A^{-1} - \frac{A^{-1}uu^\top A^{-1}}{1 + u^\top A^{-1}u}. \tag{35}$$

Because $A^{-1} = \Delta_t^{-1} I$,

$$A^{-1}u = \Delta_t^{-1}\sqrt{\beta_t}\, v, \qquad u^\top A^{-1}u = \frac{\beta_t}{\Delta_t}\|v\|^2.$$

Also,

$$A^{-1}uu^\top A^{-1} = \frac{\beta_t}{\Delta_t^2} vv^\top.$$

Substitute into (35):

$$(\Sigma_t^{\det})^{-1} = \frac{1}{\Delta_t} I - \frac{(\beta_t/\Delta_t^2)vv^\top}{1 + (\beta_t/\Delta_t)\|v\|^2}.$$

Multiply numerator and denominator of the fraction by $\Delta_t$:

$$Q_t^{\det} = \frac{1}{\Delta_t} I - \frac{\beta_t}{\Delta_t(\Delta_t + \beta_t\|v\|^2)} vv^\top. \tag{36}$$

This matrix is dense for every $t > 0$, because $vv^\top$ is dense. The corrected interpretation is:

> The deterministic control shows that a dense noised precision can arise from a rank-one propagation mechanism. This is different from the specific Gaussian-Markov band fill-in mechanism of the stochastic AR(1) chain.

# 3 Laplace $K = 1$: exact characteristic function

## 3.1 Clean characteristic function

Let $\xi = (\xi_0, \xi_1) \in \mathbb{R}^2$. By definition,

$$\Phi_0(\xi_0, \xi_1) = \mathbb{E}\Big[e^{i(\xi_0 A_0 + \xi_1 A_1)}\Big].$$

Since $A_1 = \alpha A_0 + \varepsilon$,

$$\xi_0 A_0 + \xi_1 A_1 = (\xi_0 + \alpha\xi_1)A_0 + \xi_1\varepsilon.$$

Because $A_0$ and $\varepsilon$ are independent,

$$\Phi_0(\xi_0, \xi_1) = \mathbb{E}\Big[e^{i(\xi_0 + \alpha\xi_1)A_0}\Big] \cdot \mathbb{E}\Big[e^{i\xi_1\varepsilon}\Big]. \tag{37}$$

**Gaussian factor.** If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}[e^{isY}] = \exp\Big(is\mu - \frac{1}{2}\sigma^2 s^2\Big).$$

Apply this with $Y = A_0$ and $s = \xi_0 + \alpha\xi_1$:

$$\mathbb{E}\Big[e^{i(\xi_0 + \alpha\xi_1)A_0}\Big] = \exp\Big(i\mu_0(\xi_0 + \alpha\xi_1) - \frac{1}{2}\sigma_0^2(\xi_0 + \alpha\xi_1)^2\Big). \tag{38}$$

**Laplace factor.** We compute the characteristic function of $\varepsilon \sim \text{Laplace}(0, b)$:

$$\begin{aligned}\mathbb{E}[e^{is\varepsilon}] &= \int_{-\infty}^{\infty} e^{isu}\frac{e^{-|u|/b}}{2b}\,du \\ &= \frac{1}{2b}\int_{-\infty}^{0} e^{isu}e^{u/b}\,du + \frac{1}{2b}\int_{0}^{\infty} e^{isu}e^{-u/b}\,du. \end{aligned} \tag{39}$$

The first integral is

$$\int_{-\infty}^{0} e^{(is+1/b)u}\,du = \left[\frac{e^{(is+1/b)u}}{is+1/b}\right]_{u=-\infty}^{u=0} = \frac{1}{is+1/b},$$

because $\Re(is + 1/b) = 1/b > 0$, so the exponential vanishes at $-\infty$.

The second integral is

$$\int_{0}^{\infty} e^{(is-1/b)u}\,du = \left[\frac{e^{(is-1/b)u}}{is-1/b}\right]_{u=0}^{u=\infty} = -\frac{1}{is-1/b} = \frac{1}{1/b-is}.$$

Hence

$$\mathbb{E}[e^{is\varepsilon}] = \frac{1}{2b}\left(\frac{1}{is+1/b} + \frac{1}{1/b-is}\right).$$

Put the two fractions over a common denominator:

$$\frac{1}{is+1/b} + \frac{1}{1/b-is} = \frac{(1/b-is)+(is+1/b)}{(is+1/b)(1/b-is)} = \frac{2/b}{1/b^2+s^2}.$$

Therefore

$$\mathbb{E}[e^{is\varepsilon}] = \frac{1}{1+b^2s^2}. \tag{40}$$

Combining (37), (38), and (40),

$$\Phi_0(\xi_0, \xi_1) = \exp\Big(i\mu_0(\xi_0+\alpha\xi_1) - \frac{1}{2}\sigma_0^2(\xi_0+\alpha\xi_1)^2\Big)\frac{1}{1+b^2\xi_1^2}. \tag{41}$$

### 3.2 Characteristic function after OU noising

Conditionally on $A = (A_0, A_1)$, one has

$$X_t = e^{-t}A + \sqrt{\Delta_t}Z.$$

Hence

$$\begin{aligned}\mathbb{E}[e^{i\xi^\top X_t} \mid A] &= \mathbb{E}[e^{i\xi^\top(e^{-t}A+\sqrt{\Delta_t}Z)} \mid A] \\ &= e^{ie^{-t}\xi^\top A}\,\mathbb{E}[e^{i\sqrt{\Delta_t}\,\xi^\top Z}] \\ &= e^{ie^{-t}\xi^\top A}\,e^{-\Delta_t\|\xi\|^2/2}. \end{aligned} \tag{42}$$

Taking expectation in $A$,

$$\Phi_t(\xi) = e^{-\Delta_t\|\xi\|^2/2}\,\Phi_0(e^{-t}\xi). \tag{43}$$

Insert (41):

$$\Phi_t(\xi_0, \xi_1) = \exp\Big(ie^{-t}\mu_0(\xi_0+\alpha\xi_1) - \frac{1}{2}\Delta_t(\xi_0^2+\xi_1^2) - \frac{1}{2}e^{-2t}\sigma_0^2(\xi_0+\alpha\xi_1)^2\Big)\frac{1}{1+b^2e^{-2t}\xi_1^2}. \tag{44}$$

A centered Gaussian characteristic function is always of the form $\exp(-\xi^\top\Sigma\xi/2)$, possibly multiplied by a pure phase $e^{im^\top\xi}$ if the mean is not zero. The extra factor

$$\frac{1}{1+b^2e^{-2t}\xi_1^2}$$

in (44) is rational, not exponential-quadratic. Therefore the diffused Laplace law is not Gaussian for any finite $t$.

## 4 Derivation of the normal-Laplace convolution

Fix $\beta > 0$ and $\sigma^2 > 0$. Define

$$h(y;\beta,\sigma^2) = \int_{-\infty}^{\infty} \frac{e^{-|u|/\beta}}{2\beta} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-u)^2}{2\sigma^2}\right) du. \tag{45}$$

This is the density of $L+G$ with $L \sim \text{Laplace}(0,\beta)$ and $G \sim \mathcal{N}(0,\sigma^2)$ independent.

### 4.1 Split the integral at $u = 0$

Write

$$h(y;\beta,\sigma^2) = \frac{1}{2\beta\sqrt{2\pi\sigma^2}} \Big(I_+(y) + I_-(y)\Big), \tag{46}$$

where

$$I_+(y) = \int_0^{\infty} \exp\left(-\frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2}\right) du, \tag{47}$$

$$I_-(y) = \int_{-\infty}^{0} \exp\left(\frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2}\right) du. \tag{48}$$

### 4.2 Complete the square in $I_+$

Start with the exponent:

$$\begin{aligned}
-\frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2} &= -\frac{u}{\beta} - \frac{u^2 - 2yu + y^2}{2\sigma^2} \\
&= -\frac{1}{2\sigma^2}\left(u^2 - 2yu + y^2 + \frac{2\sigma^2}{\beta}u\right) \\
&= -\frac{1}{2\sigma^2}\left(u^2 - 2\left(y - \frac{\sigma^2}{\beta}\right)u + y^2\right).
\end{aligned} \tag{49}$$

Now insert and subtract $\left(y - \sigma^2/\beta\right)^2$:

$$\begin{aligned}
u^2 - 2\left(y - \frac{\sigma^2}{\beta}\right)u + y^2 &= \left(u - \left(y - \frac{\sigma^2}{\beta}\right)\right)^2 + y^2 - \left(y - \frac{\sigma^2}{\beta}\right)^2 \\
&= \left(u - \left(y - \frac{\sigma^2}{\beta}\right)\right)^2 + \left(y^2 - y^2 + \frac{2y\sigma^2}{\beta} - \frac{\sigma^4}{\beta^2}\right) \\
&= \left(u - \left(y - \frac{\sigma^2}{\beta}\right)\right)^2 + \frac{2y\sigma^2}{\beta} - \frac{\sigma^4}{\beta^2}.
\end{aligned} \tag{50}$$

Substitute (50) into (49):

$$-\frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2} = -\frac{\left(u - (y - \sigma^2/\beta)\right)^2}{2\sigma^2} - \frac{y}{\beta} + \frac{\sigma^2}{2\beta^2}. \tag{51}$$

Therefore

$$I_+(y) = e^{-y/\beta + \sigma^2/(2\beta^2)} \int_0^{\infty} \exp\left(-\frac{(u - (y - \sigma^2/\beta))^2}{2\sigma^2}\right) du. \tag{52}$$

Set

$$z = \frac{u - (y - \sigma^2/\beta)}{\sigma}, \qquad du = \sigma dz.$$

When $u = 0$,

$$z = -\frac{y}{\sigma} + \frac{\sigma}{\beta},$$

and when $u \to \infty$, $z \to \infty$. Hence

$$\begin{aligned} I_+(y) &= e^{-y/\beta+\sigma^2/(2\beta^2)}\sigma \int_{-y/\sigma+\sigma/\beta}^{\infty} e^{-z^2/2} dz \\ &= e^{-y/\beta+\sigma^2/(2\beta^2)}\sigma\sqrt{2\pi}\,\Phi\left(\frac{y}{\sigma} - \frac{\sigma}{\beta}\right). \end{aligned} \tag{53}$$

### 4.3 Complete the square in $I_-$

Now start from

$$\begin{aligned} \frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2} &= \frac{u}{\beta} - \frac{u^2 - 2yu + y^2}{2\sigma^2} \\ &= -\frac{1}{2\sigma^2}\left(u^2 - 2yu + y^2 - \frac{2\sigma^2}{\beta}u\right) \\ &= -\frac{1}{2\sigma^2}\left(u^2 - 2\left(y + \frac{\sigma^2}{\beta}\right)u + y^2\right). \end{aligned} \tag{54}$$

Insert and subtract $\left(y + \sigma^2/\beta\right)^2$:

$$\begin{aligned} u^2 - 2\left(y + \frac{\sigma^2}{\beta}\right)u + y^2 &= \left(u - \left(y + \frac{\sigma^2}{\beta}\right)\right)^2 + y^2 - \left(y + \frac{\sigma^2}{\beta}\right)^2 \\ &= \left(u - \left(y + \frac{\sigma^2}{\beta}\right)\right)^2 - \frac{2y\sigma^2}{\beta} - \frac{\sigma^4}{\beta^2}. \end{aligned} \tag{55}$$

Substitute into (54):

$$\frac{u}{\beta} - \frac{(y-u)^2}{2\sigma^2} = -\frac{\left(u - (y + \sigma^2/\beta)\right)^2}{2\sigma^2} + \frac{y}{\beta} + \frac{\sigma^2}{2\beta^2}. \tag{56}$$

Hence

$$I_-(y) = e^{y/\beta+\sigma^2/(2\beta^2)} \int_{-\infty}^{0} \exp\left(-\frac{(u - (y + \sigma^2/\beta))^2}{2\sigma^2}\right) du. \tag{57}$$

Set

$$z = \frac{u - (y + \sigma^2/\beta)}{\sigma}, \qquad du = \sigma dz.$$

When $u = 0$,

$$z = -\frac{y}{\sigma} - \frac{\sigma}{\beta},$$

and when $u \to -\infty$, $z \to -\infty$. Therefore

$$\begin{aligned} I_-(y) &= e^{y/\beta+\sigma^2/(2\beta^2)}\sigma \int_{-\infty}^{-y/\sigma-\sigma/\beta} e^{-z^2/2} dz \\ &= e^{y/\beta+\sigma^2/(2\beta^2)}\sigma\sqrt{2\pi}\,\Phi\left(-\frac{y}{\sigma} - \frac{\sigma}{\beta}\right). \end{aligned} \tag{58}$$

### 4.4 Final formula and an equivalent erfc form

Insert (53) and (58) into (46):

$$h(y;\beta,\sigma^2) = \frac{e^{\sigma^2/(2\beta^2)}}{2\beta}\left[e^{-y/\beta}\Phi\left(\frac{y}{\sigma}-\frac{\sigma}{\beta}\right)+e^{y/\beta}\Phi\left(-\frac{y}{\sigma}-\frac{\sigma}{\beta}\right)\right]. \tag{59}$$

Using $\Phi(z)=\frac{1}{2}\operatorname{erfc}(-z/\sqrt{2})$, one also gets

$$h(y;\beta,\sigma^2) = \frac{e^{\sigma^2/(2\beta^2)}}{4\beta}\left[e^{-y/\beta}\operatorname{erfc}\left(\frac{\sigma^2/\beta-y}{\sqrt{2}\sigma}\right)+e^{y/\beta}\operatorname{erfc}\left(\frac{\sigma^2/\beta+y}{\sqrt{2}\sigma}\right)\right]. \tag{60}$$

This is the corrected formula used in the new code.

### 4.5 Derivative formulas for $h$

Set

$$u=\frac{y}{\sigma}-\frac{\sigma}{\beta},\qquad v=-\frac{y}{\sigma}-\frac{\sigma}{\beta},\qquad C=\frac{e^{\sigma^2/(2\beta^2)}}{2\beta}.$$

Then (59) reads

$$h(y)=C\big(e^{-y/\beta}\Phi(u)+e^{y/\beta}\Phi(v)\big).$$

Differentiate the first term:

$$\begin{aligned}\frac{d}{dy}\left[e^{-y/\beta}\Phi(u)\right] &= -\frac{1}{\beta}e^{-y/\beta}\Phi(u)+e^{-y/\beta}\phi(u)\frac{du}{dy}\\ &= -\frac{1}{\beta}e^{-y/\beta}\Phi(u)+\frac{1}{\sigma}e^{-y/\beta}\phi(u),\end{aligned} \tag{61}$$

where $\phi(z)=\frac{1}{\sqrt{2\pi}}e^{-z^2/2}$. Differentiate the second term:

$$\begin{aligned}\frac{d}{dy}\left[e^{y/\beta}\Phi(v)\right] &= \frac{1}{\beta}e^{y/\beta}\Phi(v)+e^{y/\beta}\phi(v)\frac{dv}{dy}\\ &= \frac{1}{\beta}e^{y/\beta}\Phi(v)-\frac{1}{\sigma}e^{y/\beta}\phi(v).\end{aligned} \tag{62}$$

We now show that the Gaussian-density pieces cancel. Compute

$$\begin{aligned}e^{-y/\beta}\phi(u) &= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y}{\beta}-\frac{1}{2}\left(\frac{y}{\sigma}-\frac{\sigma}{\beta}\right)^2\right)\\ &= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y}{\beta}-\frac{y^2}{2\sigma^2}+\frac{y}{\beta}-\frac{\sigma^2}{2\beta^2}\right)\\ &= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y^2}{2\sigma^2}-\frac{\sigma^2}{2\beta^2}\right).\end{aligned} \tag{63}$$

Similarly,

$$\begin{aligned}e^{y/\beta}\phi(v) &= \frac{1}{\sqrt{2\pi}}\exp\left(\frac{y}{\beta}-\frac{1}{2}\left(-\frac{y}{\sigma}-\frac{\sigma}{\beta}\right)^2\right)\\ &= \frac{1}{\sqrt{2\pi}}\exp\left(\frac{y}{\beta}-\frac{y^2}{2\sigma^2}-\frac{y}{\beta}-\frac{\sigma^2}{2\beta^2}\right)\\ &= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y^2}{2\sigma^2}-\frac{\sigma^2}{2\beta^2}\right).\end{aligned} \tag{64}$$

Therefore the $\sigma^{-1}$ terms in (61) and (62) cancel. What remains is

$$h'(y;\beta,\sigma^2)=\frac{e^{\sigma^2/(2\beta^2)}}{2\beta^2}\left[e^{y/\beta}\Phi(v)-e^{-y/\beta}\Phi(u)\right]. \tag{65}$$

**Second derivative from an ODE identity.** The centered Laplace density $\ell_\beta(y) = \frac{1}{2\beta}e^{-|y|/\beta}$ has Fourier transform

$$\widehat{\ell_\beta}(\xi) = \frac{1}{1+\beta^2\xi^2}.$$

Therefore

$$(1+\beta^2\xi^2)\widehat{\ell_\beta}(\xi) = 1.$$

Under the Fourier transform, multiplication by $1+\beta^2\xi^2$ corresponds to the operator $1-\beta^2 d^2/dy^2$. Hence, in the distributional sense,

$$\left(1-\beta^2\frac{d^2}{dy^2}\right)\ell_\beta = \delta_0. \tag{66}$$

Now $h = \ell_\beta * \phi_{\sigma^2}$. Convolving (66) with $\phi_{\sigma^2}$,

$$\left(1-\beta^2\frac{d^2}{dy^2}\right)h = \phi_{\sigma^2}.$$

Therefore

$$h''(y;\beta,\sigma^2) = \frac{h(y;\beta,\sigma^2)-\phi_{\sigma^2}(y)}{\beta^2}. \tag{67}$$

# 5 Posterior of $A_0$ given $X_0 = x_0$ and exact density reduction

From (2),

$$X_0 = e^{-t}A_0 + \sqrt{\Delta_t}Z_0.$$

Hence

$$X_0 \mid A_0 = a_0 \sim \mathcal{N}(e^{-t}a_0, \Delta_t).$$

The posterior density is proportional to prior times likelihood:

$$p(a_0 \mid x_0) \propto \exp\left(-\frac{(a_0-\mu_0)^2}{2\sigma_0^2} - \frac{(x_0-e^{-t}a_0)^2}{2\Delta_t}\right). \tag{68}$$

## 5.1 Complete the square explicitly

Expand both squares:

$$-\frac{(a_0-\mu_0)^2}{2\sigma_0^2} = -\frac{a_0^2}{2\sigma_0^2} + \frac{\mu_0}{\sigma_0^2}a_0 - \frac{\mu_0^2}{2\sigma_0^2}, \tag{69}$$

$$-\frac{(x_0-e^{-t}a_0)^2}{2\Delta_t} = -\frac{x_0^2}{2\Delta_t} + \frac{e^{-t}x_0}{\Delta_t}a_0 - \frac{e^{-2t}}{2\Delta_t}a_0^2. \tag{70}$$

Add (69) and (70):

$$\begin{aligned}
&-\frac{(a_0-\mu_0)^2}{2\sigma_0^2} - \frac{(x_0-e^{-t}a_0)^2}{2\Delta_t} \\
&= -\frac{1}{2}\left(\sigma_0^{-2} + e^{-2t}\Delta_t^{-1}\right)a_0^2 + \left(\mu_0\sigma_0^{-2} + e^{-t}x_0\Delta_t^{-1}\right)a_0 + \text{constant in } a_0.
\end{aligned} \tag{71}$$

Define

$$v_t^{-1} := \sigma_0^{-2} + e^{-2t}\Delta_t^{-1}, \qquad m_t(x_0) := v_t\left(\frac{\mu_0}{\sigma_0^2} + \frac{e^{-t}x_0}{\Delta_t}\right). \tag{72}$$

Then (71) becomes
$$-\frac{1}{2v_t}a_0^2 + \frac{m_t(x_0)}{v_t}a_0 + \text{constant}.$$
Complete the square:
$$\begin{aligned} -\frac{1}{2v_t}a_0^2 + \frac{m_t}{v_t}a_0 &= -\frac{1}{2v_t}\left(a_0^2 - 2m_t a_0\right) \\ &= -\frac{1}{2v_t}\left((a_0 - m_t)^2 - m_t^2\right) \\ &= -\frac{(a_0 - m_t)^2}{2v_t} + \frac{m_t^2}{2v_t}. \end{aligned} \tag{73}$$
Hence
$$A_0 \mid X_0 = x_0 \sim \mathcal{N}(m_t(x_0), v_t). \tag{74}$$
Since $v_t$ depends only on $t$, not on $x_0$,
$$m_t'(x_0) = \frac{v_t e^{-t}}{\Delta_t}. \tag{75}$$

**Marginal law of $X_0$.** Directly from (2),
$$X_0 = e^{-t}A_0 + \sqrt{\Delta_t}Z_0$$
is a sum of independent Gaussian variables, so
$$X_0 \sim \mathcal{N}(e^{-t}\mu_0, v_x), \qquad v_x := e^{-2t}\sigma_0^2 + \Delta_t. \tag{76}$$

## 5.2 Exact density reduction

Start from the full joint integral over $a_0$ and $\varepsilon$:
$$p_t(x_0, x_1) = \int_{\mathbb{R}}\int_{\mathbb{R}} \phi_{\sigma_0^2}(a_0 - \mu_0) \cdot \frac{e^{-|\varepsilon|/b}}{2b} \cdot \phi_{\Delta_t}(x_0 - e^{-t}a_0) \cdot \phi_{\Delta_t}(x_1 - e^{-t}(\alpha a_0 + \varepsilon))\, d\varepsilon\, da_0. \tag{77}$$
Fix $a_0$ and define
$$y = x_1 - e^{-t}\alpha a_0.$$
Then
$$x_1 - e^{-t}(\alpha a_0 + \varepsilon) = y - e^{-t}\varepsilon.$$
The random variable $e^{-t}\varepsilon$ has Laplace law with scale $\beta_t = e^{-t}b$, so the $\varepsilon$-integral in (77) is exactly
$$h(y; \beta_t, \Delta_t) = h(x_1 - e^{-t}\alpha a_0; \beta_t, \Delta_t). \tag{78}$$
Therefore (77) becomes
$$p_t(x_0, x_1) = \int_{\mathbb{R}} \phi_{\sigma_0^2}(a_0 - \mu_0)\, \phi_{\Delta_t}(x_0 - e^{-t}a_0)\, h(x_1 - c_t a_0; \beta_t, \Delta_t)\, da_0, \qquad c_t = e^{-t}\alpha. \tag{79}$$
By Bayes' rule,
$$\phi_{\sigma_0^2}(a_0 - \mu_0)\, \phi_{\Delta_t}(x_0 - e^{-t}a_0) = p(x_0)\, p(a_0 \mid x_0).$$
We already computed both factors:
$$p(x_0) = \phi_{v_x}(x_0 - e^{-t}\mu_0), \qquad p(a_0 \mid x_0) = \phi_{v_t}(a_0 - m_t(x_0)).$$
Substitute these into (79):
$$\begin{aligned} p_t(x_0, x_1) &= \phi_{v_x}(x_0 - e^{-t}\mu_0) \int_{\mathbb{R}} \phi_{v_t}(a_0 - m_t(x_0))\, h(x_1 - c_t a_0; \beta_t, \Delta_t)\, da_0 \\ &= \phi_{v_x}(x_0 - e^{-t}\mu_0)\, \mathbb{E}[h(x_1 - c_t A_0; \beta_t, \Delta_t) \,|\, X_0 = x_0]\,. \end{aligned} \tag{80}$$
This is the exact one-dimensional reduction used in the short note.

# 6 Exact score and exact effective precision

Define, for $r = 0, 1, 2$,

$$I_r(x_0, x_1; t) := \mathbb{E}\Big[h^{(r)}(x_1 - c_t A_0; \beta_t, \Delta_t) \,\Big|\, X_0 = x_0\Big]. \tag{81}$$

Then (80) reads

$$p_t(x_0, x_1) = \phi_{v_x}(x_0 - e^{-t}\mu_0) I_0(x_0, x_1; t). \tag{82}$$

## 6.1 Derivative rule for $I_r$ with respect to $x_0$

Write the posterior variable as

$$A_0 = m_t(x_0) + \sqrt{v_t} Z, \qquad Z \sim \mathcal{N}(0, 1).$$

Then

$$I_r(x_0, x_1; t) = \mathbb{E}\Big[h^{(r)}(x_1 - c_t(m_t(x_0) + \sqrt{v_t} Z))\Big].$$

Because $v_t$ is independent of $x_0$, differentiation gives

$$\begin{aligned} \partial_{x_0} I_r &= \mathbb{E}\Big[h^{(r+1)}(x_1 - c_t(m_t + \sqrt{v_t} Z)) \cdot (-c_t m_t'(x_0))\Big] \\ &= -c_t m_t'(x_0) I_{r+1}(x_0, x_1; t). \end{aligned} \tag{83}$$

This identity is the key input for all score and Hessian calculations.

## 6.2 Exact score formulas

Take logarithms in (82):

$$\log p_t(x_0, x_1) = \log \phi_{v_x}(x_0 - e^{-t}\mu_0) + \log I_0(x_0, x_1; t). \tag{84}$$

**Derivative with respect to $x_1$.** Since the Gaussian prefactor depends only on $x_0$,

$$\partial_{x_1} \log p_t(x_0, x_1) = \frac{\partial_{x_1} I_0}{I_0} = \frac{I_1}{I_0}. \tag{85}$$

**Derivative with respect to $x_0$.** First,

$$\log \phi_{v_x}(x_0 - e^{-t}\mu_0) = -\frac{1}{2}\log(2\pi v_x) - \frac{(x_0 - e^{-t}\mu_0)^2}{2v_x}.$$

Differentiate:

$$\partial_{x_0} \log \phi_{v_x}(x_0 - e^{-t}\mu_0) = -\frac{x_0 - e^{-t}\mu_0}{v_x}. \tag{86}$$

Also, by (83) with $r = 0$,

$$\partial_{x_0} I_0 = -c_t m_t'(x_0) I_1.$$

Hence

$$\partial_{x_0} \log p_t(x_0, x_1) = -\frac{x_0 - e^{-t}\mu_0}{v_x} - c_t m_t'(x_0)\frac{I_1}{I_0}. \tag{87}$$

Equations (85) and (87) are the exact score formulas used in the short note.

### 6.3 Exact Hessian formulas

Define the effective precision

$$\mathcal{H}_t(x) := -\nabla_x^2 \log p_t(x). \tag{88}$$

We now compute its entries explicitly.

**The $(1,1)$ entry.** Differentiate (85) with respect to $x_1$:

$$\begin{aligned}
\partial_{x_1x_1} \log p_t &= \partial_{x_1}\left(\frac{I_1}{I_0}\right) \\
&= \frac{(\partial_{x_1} I_1) I_0 - I_1(\partial_{x_1} I_0)}{I_0^2} \\
&= \frac{I_2 I_0 - I_1^2}{I_0^2}.
\end{aligned} \tag{89}$$

Therefore

$$(\mathcal{H}_t(x))_{11} = -\partial_{x_1x_1} \log p_t(x) = \frac{I_1^2 - I_0 I_2}{I_0^2}. \tag{90}$$

**The mixed entry.** Differentiate (85) with respect to $x_0$ and use (83) twice:

$$\begin{aligned}
\partial_{x_0x_1} \log p_t &= \partial_{x_0}\left(\frac{I_1}{I_0}\right) \\
&= \frac{(\partial_{x_0} I_1) I_0 - I_1(\partial_{x_0} I_0)}{I_0^2} \\
&= \frac{(-c_t m_t' I_2) I_0 - I_1(-c_t m_t' I_1)}{I_0^2} \\
&= -c_t m_t'(x_0) \frac{I_2 I_0 - I_1^2}{I_0^2} \\
&= c_t m_t'(x_0)(\mathcal{H}_t(x))_{11}.
\end{aligned} \tag{91}$$

Therefore

$$(\mathcal{H}_t(x))_{01} = -\partial_{x_0x_1} \log p_t(x) = -c_t m_t'(x_0)(\mathcal{H}_t(x))_{11}. \tag{92}$$

**The $(0,0)$ entry.** Differentiate (87) with respect to $x_0$:

$$\partial_{x_0x_0} \log p_t = -\frac{1}{v_x} - c_t m_t'(x_0) \partial_{x_0}\left(\frac{I_1}{I_0}\right) \tag{93}$$

because $m_t'(x_0)$ is constant in $x_0$. Using (91),

$$\partial_{x_0}\left(\frac{I_1}{I_0}\right) = c_t m_t'(x_0)(\mathcal{H}_t(x))_{11}.$$

Substitute into (93):

$$\partial_{x_0x_0} \log p_t = -\frac{1}{v_x} - c_t^2 \big(m_t'(x_0)\big)^2 (\mathcal{H}_t(x))_{11}. \tag{94}$$

Therefore

$$(\mathcal{H}_t(x))_{00} = \frac{1}{v_x} + c_t^2 \big(m_t'(x_0)\big)^2 (\mathcal{H}_t(x))_{11}. \tag{95}$$

This completes the derivation of the exact curvature matrix.

### 6.4 Why constant Hessian would force Gaussianity

The next statement is the clean logical bridge from the formulas above to the presentation claim.

**Proposition 1.** *Let $p$ be a strictly positive $C^2$ density on $\mathbb{R}^n$. If $-\nabla^2 \log p(x) \equiv H$ is constant, then $p$ is Gaussian.*

*Proof.* If $-\nabla^2 \log p(x) \equiv H$, then

$$\nabla^2 \log p(x) = -H \qquad \text{for all } x.$$

Integrating once, there exists a vector $b \in \mathbb{R}^n$ such that

$$\nabla \log p(x) = -Hx + b.$$

Now integrate along the straight segment $s \mapsto sx$ from 0 to $x$:

$$\begin{aligned}
\log p(x) - \log p(0) &= \int_0^1 \frac{d}{ds} \log p(sx)\, ds \\
&= \int_0^1 \langle \nabla \log p(sx), x \rangle\, ds \\
&= \int_0^1 \langle -H(sx) + b, x \rangle\, ds \\
&= -\int_0^1 s\, x^\top Hx\, ds + \int_0^1 b^\top x\, ds \\
&= -\frac{1}{2} x^\top Hx + b^\top x.
\end{aligned} \tag{96}$$

Hence

$$p(x) = C \exp\Big( -\frac{1}{2} x^\top Hx + b^\top x \Big)$$

for some constant $C > 0$. Completing the square shows that this is a Gaussian density. □

Apply the proposition to $p = p_t$. Since the characteristic function in (44) is not Gaussian for finite $t$, the Hessian cannot be constant. Therefore the exact formulas (90)–(95) are not merely decorative: they encode a genuinely non-Gaussian effect.

## 7 What can be stated exactly for $K = 2$

Let

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \qquad A_1 = \alpha A_0 + \varepsilon_1, \qquad A_2 = \alpha A_1 + \varepsilon_2,$$

with $\varepsilon_1, \varepsilon_2 \overset{\text{iid}}{\sim} \text{Laplace}(0, b)$, followed by the same OU noising on each coordinate. Then the exact noisy density is

$$p_t(x_0, x_1, x_2) = \int_{\mathbb{R}} \phi_{\sigma_0^2}(a_0 - \mu_0) \phi_{\Delta_t}(x_0 - e^{-t} a_0)\; J_t(a_0; x_1, x_2)\, da_0, \tag{97}$$

where

$$J_t(a_0; x_1, x_2) = \int_{\mathbb{R}} \frac{e^{-|a_1 - \alpha a_0|/b}}{2b}\, \phi_{\Delta_t}(x_1 - e^{-t} a_1)\, h(x_2 - e^{-t} \alpha a_1; \beta_t, \Delta_t)\, da_1. \tag{98}$$

Formulas (97)–(98) are exact. However, this is still a nested two-layer integral representation. The current project does *not* claim an elementary closed form comparable to the $K = 1$ formula (80). That stronger claim would require additional work.

## 8 Numerical realization used in the figures

The final figures do not use Monte Carlo. They evaluate the exact formulas above by Gauss-Hermite quadrature.

If $A \sim \mathcal{N}(m, v)$ and $f$ is integrable, then

$$\mathbb{E}[f(A)] = \int_{\mathbb{R}} f(a) \frac{1}{\sqrt{2\pi v}} e^{-(a-m)^2/(2v)} da. \tag{99}$$

Set

$$a = m + \sqrt{2v}\, z, \qquad da = \sqrt{2v}\, dz.$$

Then

$$\begin{aligned} \mathbb{E}[f(A)] &= \int_{\mathbb{R}} f(m + \sqrt{2v} z) \frac{1}{\sqrt{2\pi v}} e^{-z^2} \sqrt{2v}\, dz \\ &= \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} f(m + \sqrt{2v} z) e^{-z^2} dz. \end{aligned} \tag{100}$$

If $(z_i, w_i)_{i=1}^M$ are Hermite nodes and weights for $\int e^{-z^2} g(z) dz$, then

$$\mathbb{E}[f(A)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^{M} w_i f(m + \sqrt{2v} z_i). \tag{101}$$

This is exactly the formula used in the code to compute $I_0$, $I_1$, and $I_2$.

The code also includes independent checks:

1. direct inversion checks for the Gaussian finite-chain precision and for the deterministic control;
2. a direct convolution integral confirming the corrected normal-Laplace formula;
3. an independent double integral confirming the $K = 1$ reduction formula;
4. finite-difference checks confirming the exact score and Hessian formulas.

The script `code/run_checks.py` writes the resulting errors to `code/check_report.txt`.

## 9 Audit summary

The final short note keeps only the claims that are either exact or clearly marked.

1. **Kept exactly:**
   - the Gaussian OU identities $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ and $Q_t = \Sigma_t^{-1}$;
   - the small-time expansion $Q_t = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}$ under the convergence condition $c_t \|Q_0\| < 1$;
   - the large-time expansion $Q_t = I - e^{-2t}(\Sigma_0 - I) + O(e^{-4t})$;
   - the deterministic-control inverse formula (36);
   - the exact Laplace $K = 1$ characteristic function, density reduction, score, and effective-precision formulas.
2. **Removed or weakened:**
   - the inaccurate boundary form of the finite-chain Gaussian precision;

- the claim that the original locality observables "fully characterize locality";
- the interpretation that deterministic dense precision and stochastic Markov fill-in are the same mechanism;
- the old $K = 2$ elementary closed-form claim.

3. **Main message now supported rigorously:**

$$\begin{aligned}\text{Gaussian benchmark} &\Longrightarrow -\nabla^2 \log p_t \equiv Q_t,\\ \text{Laplace benchmark} &\Longrightarrow -\nabla^2 \log p_t(x) = \mathcal{H}_t(x) \text{ depends on } x.\end{aligned}$$

   This is the central statement that the final short presentation is built to communicate.